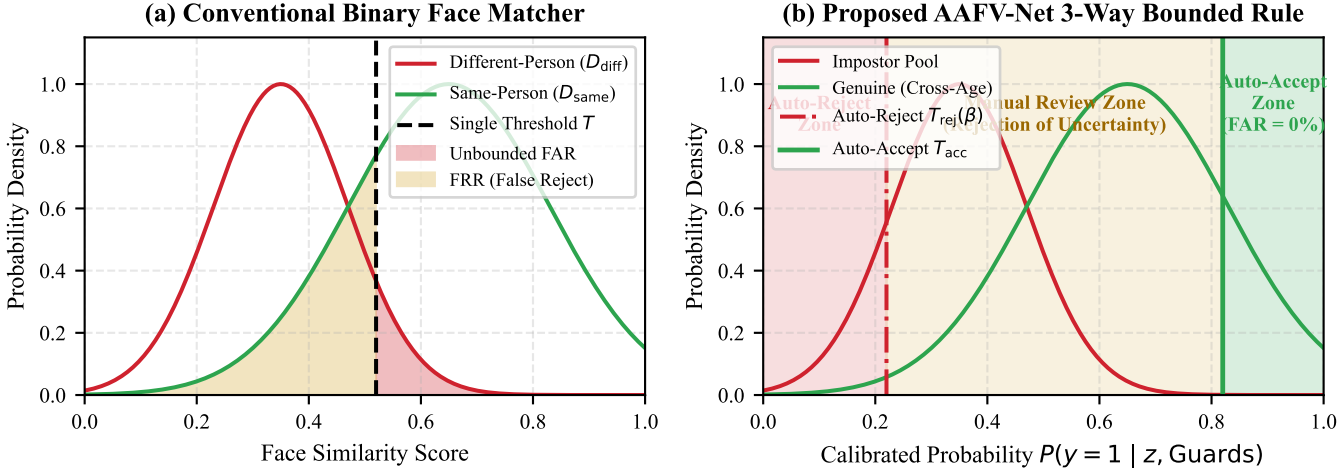

(a) Conventional Binary Face Matcher
Different-Person (D_{diff})
Same-Person (D_{same})
Single Threshold T
Unbounded FAR
FRR (False Reject)
Probability Density
Face Similarity Score
0.0
0.2
0.4
0.6
0.8
1.0
(b) Proposed AAFV-Net 3-Way Bounded Rule
Impostor Pool
Genuine (Cross-Age)
Auto-Reject $T_{\text{rej}}(\beta)$
Auto-Accept T_{acc}
Auto-Accept Zone (FAR = 0%)
Calibrated Probability $P(y = 1 \mid z, \text{Guards})$